

DCA40010

Research Methodology

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Unit I: Introduction

Meaning

Research is an endeavor to discover answers to intellectual and practical problems through the application of scientific method.

❖ “Research is a systematized effort to gain new knowledge”.

-Redman and Mory.

❖ Research is the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon about which we are concerned or interested.

Objectives of Research

The purpose of research is to discover answers through the application of scientific procedures.

❖ The objectives are:

- To gain familiarity with a phenomenon or to achieve new insights into it – Exploratory or Formulative Research.
- To portray accurately the characteristics of a particular individual, situation or a group – Descriptive Research.
- To determine the frequency with which something occurs or with which it is associated with something else – Diagnostic Research.
- To test a hypothesis of a causal relationship between variables – Hypothesis-Testing Research.

Characteristics of Research

Research is directed towards the solution of a problem.

Research is based upon observable experience or empirical evidence.

Research demands accurate observation and description.

Research involves gathering new data from primary sources or using existing data for a new purpose.

Research activities are characterized by carefully designed procedures.

Research requires expertise i.e., skill necessary to carryout investigation, search the related literature and to understand and analyze the data gathered.

Research is objective and logical – applying every possible test to validate the data collected and conclusions reached.

Research involves the quest for answers to unsolved problems.

Research requires courage.

Research is characterized by patient and unhurried activity.

Research is carefully recorded and reported.

SCIENTIFIC METHOD

- ✿ 'Science' refers to the body of systematic and organised knowledge which makes use of scientific method to acquire knowledge in a particular field of enquiry.
- ✿ Scientific method is the systematic collection of data (facts) and their theoretical treatment through proper observation, experimentation and interpretation.
- ✿ Scientific method attempts to achieve a systematic interrelation of facts by experimentation, observation, and logical arguments from accepted postulates and a combination of these three in varying proportions.

BASIC POSTULATES **IN SCIENTIFIC METHOD**

- 👉 It relies on empirical evidence.
- 👉 It utilizes relevant concepts.
- 👉 It is committed to only objective considerations.
- 👉 It presupposes ethical neutrality.
- 👉 It results into probabilistic predictions.
- 👉 The methodology is made known.
- 👉 Aims at formulating scientific theories.

CRITERIA OF A GOOD RESEARCH

- ⊕ Purpose clearly defined.
- ⊕ Research process detailed.
- ⊕ Research design thoroughly planned.
- ⊕ High ethical standards applied.
- ⊕ Limitations frankly revealed.
- ⊕ Adequate analysis for decision maker's needs.
- ⊕ Findings presented unambiguously.
- ⊕ Conclusions justified.
- ⊕ Researcher's experience reflected.

QUALITIES OF A GOOD RESEARCH

- Systematic
- Logical
- Empirical
- Replicable
- Creative
- Use of multiple methods

NEED FOR RESEARCH

- ♪ EXPLORATION
- ♪ DESCRIBE
- ♪ DIAGNOSE
- ♪ HYPOTHESIS
- ♪ INDUCTIONS AND DEDUCTIONS

- ❑ Solves investment problems
- ❑ Solves pricing problems
- ❑ Solves allocation problems
- ❑ Solves decision making issues in HR
- ❑ Solves various operational and planning problems of business and industry

- ❑ Provides the basis for all government policies in our economic system.
- ❑ Helps social scientists in studying social relationships and in seeking answers to various social problems.
- ❑ For students, research means a careerism or a way to attain a high position in the social structure.
- ❑ For professionals in research, it may mean a source of livelihood.

- ❑ For philosophers and thinkers, research means the outlet for new ideas and insights.
- ❑ For literary men and women, research means development of new styles and creative work.
- ❑ For analysts and intellectuals, research means generalizations of new theories.

PROBLEMS IN RESEARCH

- **Not similar to science**
- **Uncontrollable variables**
- **Human tendencies**
- **Time and money**
- **Lack of computerization**
- **Lack of scientific training in the methodology of research**

- According to Herbert A Simon, decision-making involves three activities:
 - 👉 Intelligence Activity - scanning the environment for identifying conditions necessary for the decision.
 - 👉 Designing Activity - identifying, developing and analyzing the alternative courses of action.
 - 👉 Choice Activity - choosing the best course of action from among the alternatives.

FACTORS THAT AFFECT MANAGERIAL DECISIONS

- ❧ INTERNAL FACTORS – factors present inside an organisation such as resources, technology, trade unions, cash flow, manpower etc.
- ❧ EXTERNAL FACTORS – factors present outside the organisation such as government policies, political factors, socio-economic factors, legal framework, geographic and cultural factors etc.
- ❧ QUANTITATIVE FACTORS – factors that can be measured in quantities such as time, resources, cost factors etc.

- QUALITATIVE FACTORS – factors that cannot be measured in quantities such as organizational cohesiveness, sense of belonging of employees, risk of technological change etc.
- UNCERTAINTY FACTORS – factors which cannot be predicted.